0226. Invert Binary Tree

Easy · Trees

leetcode.com/problems/invert-binary-tree

Problem

- Mirror a binary tree: swap the left and right child of every node, and return the root.

Intuition

- An inverted tree is a root whose children are the inverted right and inverted left subtrees: a perfect recursive definition.

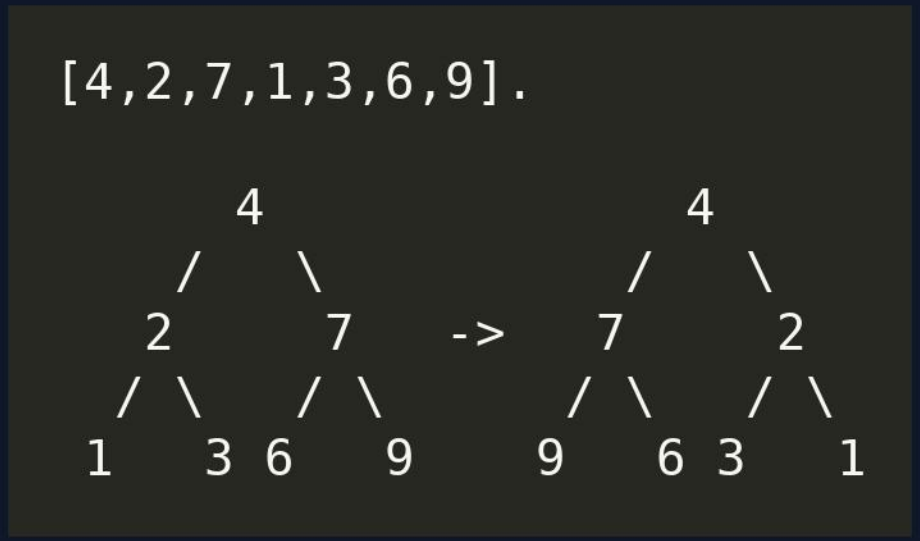
Brute Force

- There is no slower "brute force"; the only choice is recursive DFS versus iterative BFS with a queue.

Optimal Approach

- If root is null, return null.
- Swap root.left and root.right.
- Recurse on both children.

Worked Example



Java Solution

```java
public class InvertBinaryTree {
    public TreeNode invertTree(TreeNode root) {
        if (root == null) return null;
        TreeNode tmp = root.left;
        root.left = invertTree(root.right);
        root.right = invertTree(tmp);
        return root;
    }
}
```

Complexity

- Time: O(n). Space: O(h) recursion (h = height).

Pitfalls

- Swap with a temp variable, or assign from saved recursive results; never overwrite before reading.